DCF method using atomic polarization from Ground-state Alignment

Measurement of B-field strength in the ISM

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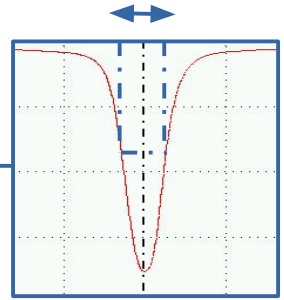
HELMHOLTZ

The Davis-Chandrasekhar-Fermi technique

The classical method (Davis 1951, Chandrasekhar & Fermi 1953)

$$B_{0,\mathrm{pos}} = \xi \sqrt{4\pi\bar{\rho}}\,\frac{\delta v_{\mathrm{los}}}{\delta\phi}$$

$\sim 0.4 - 0.6$

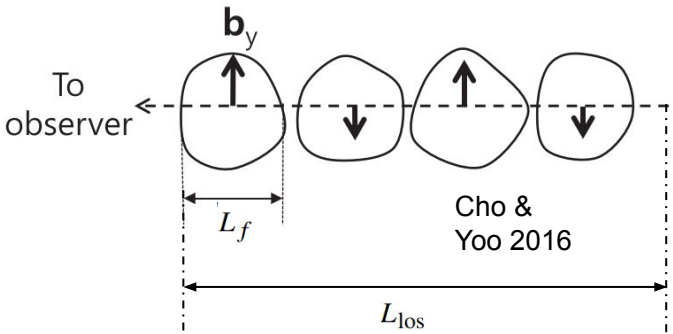


$$\phi_P = \frac{1}{2}\tan^{-1}\frac{U}{Q}$$

The classical method does not consider the effects of the driving scale of turbulence!

The Davis-Chandrasekhar-Fermi technique

The modified DCF method (Cho & Yoo 2016)



$$\delta\phi \sim \frac{\delta b_{\perp,pos}}{B_{0,pos}} \sqrt{L_f / L_{los}}$$

Modified DCF method:

$$B_{0,pos} = \xi' \sqrt{4\pi\bar{\rho}} \frac{\delta V_c}{\delta\phi}$$

$$V_{c,i} = \int v_{los} I_i(v_{los})\, dv_{los} \Big/ \int I_i(v_{los})\, dv_{los}$$

$$\sim 0.7 - 1.0$$

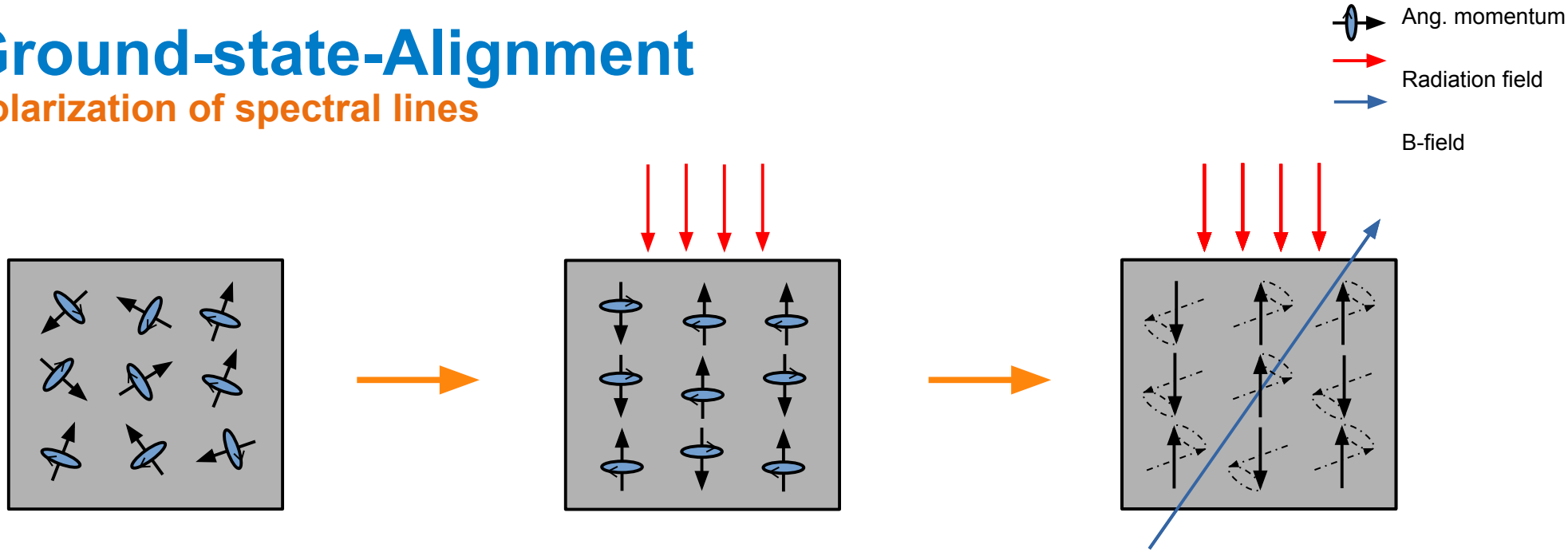
Motivation

The problem with dust polarization in DCF

- In DCF, measurement of LOS and POS components are obtained from separate targets:
  - δv from optically thin emission lines (radio)
  - $\delta \phi$ from averaged continuum (visible/IR)
- Additionally polarization efficiency can be affected by
  - Variation in grain sizes
  - Chemical composition of dust

Ground-state-Alignment

Polarization of spectral lines



Randomly oriented initial angular momenta of atoms/ions in the ISM

Anisotropic pumping causes alignment to the radiation direction

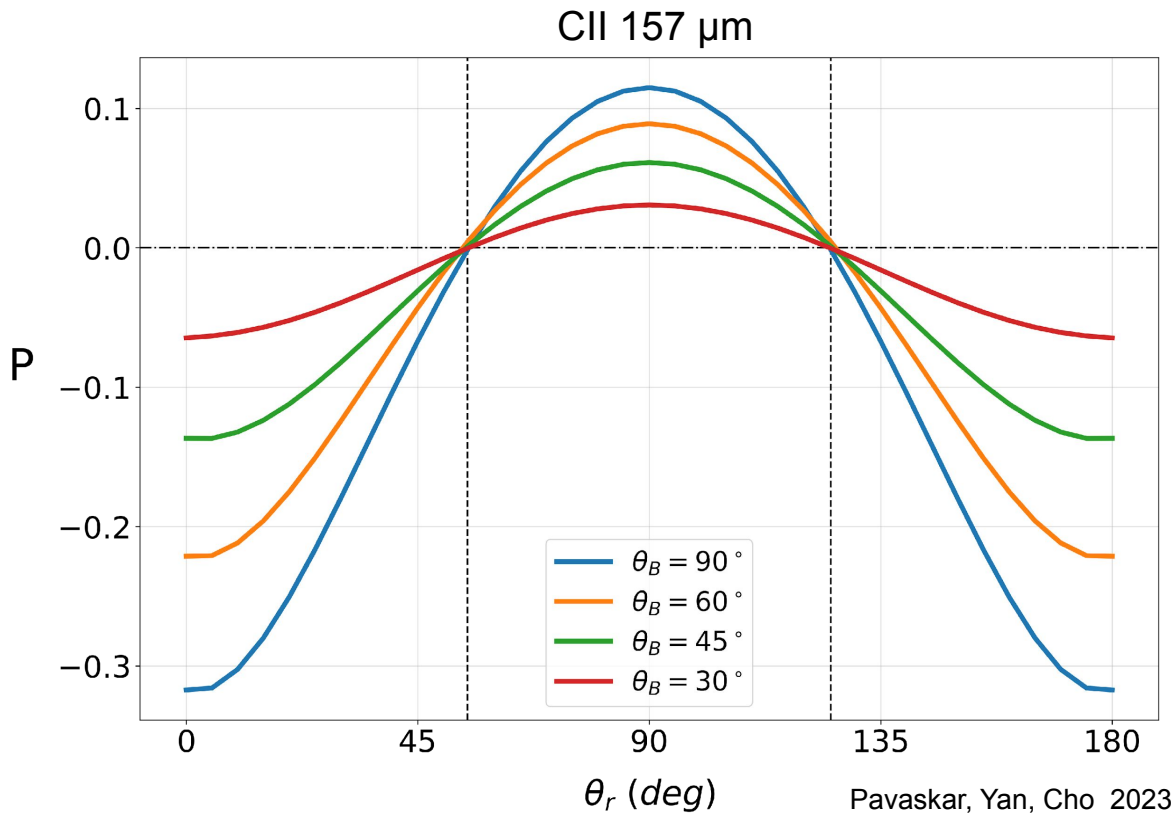
In an external B-field, gyration around the B-field direction causes realignment if $\nu_{\mathrm{L}} > R_F > \tau_c^{-1}$

Transition lines in these atoms on the ground state are polarized!

Ground-state-Alignment

Polarization of spectral lines

- Conditions for GSA effect:
  - Anisotropic radiation
  - Fine (or hyperfine) structure in the atomic ground state
  - Radiative dominant regime
  - Long lived ground (or metastable) state



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Testing DCF with GSA

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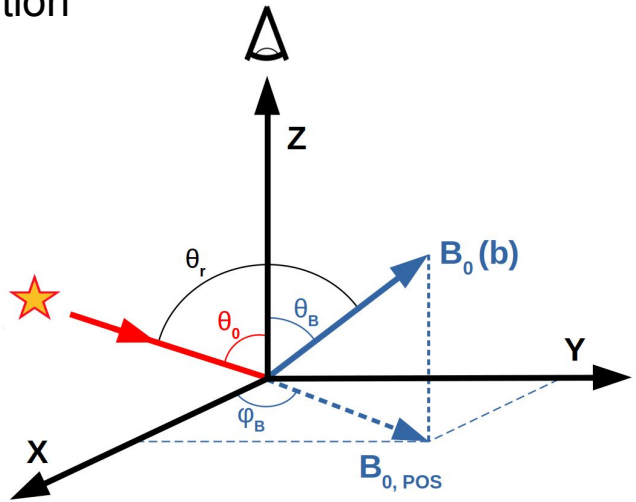
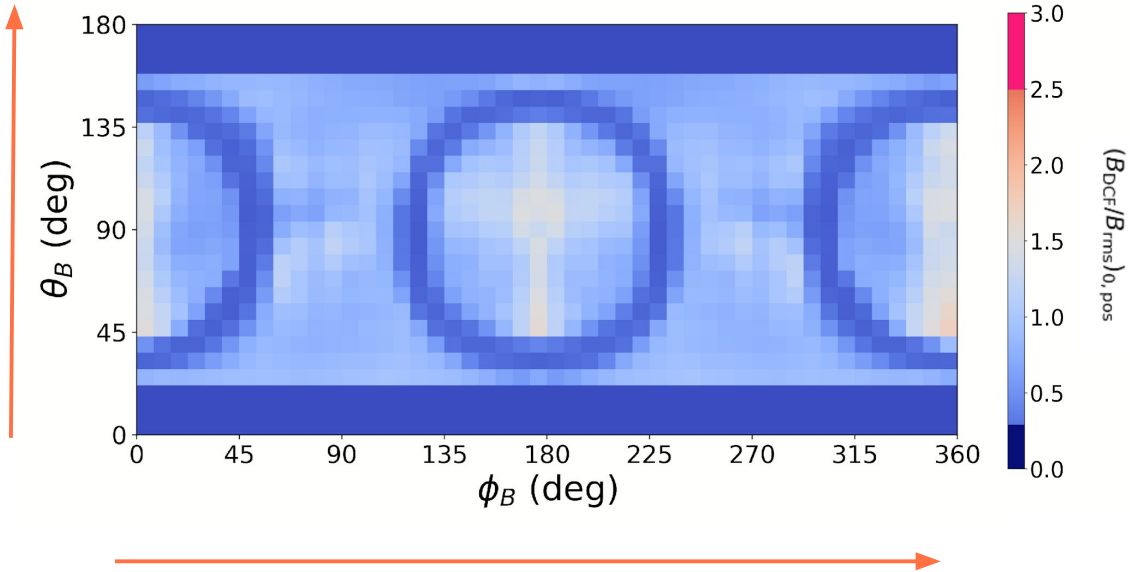
- Generate MHD simulations with diff. M_A
- Rotate the mean field vector and compute POS polarization maps
- Repeat by changing radiation source location





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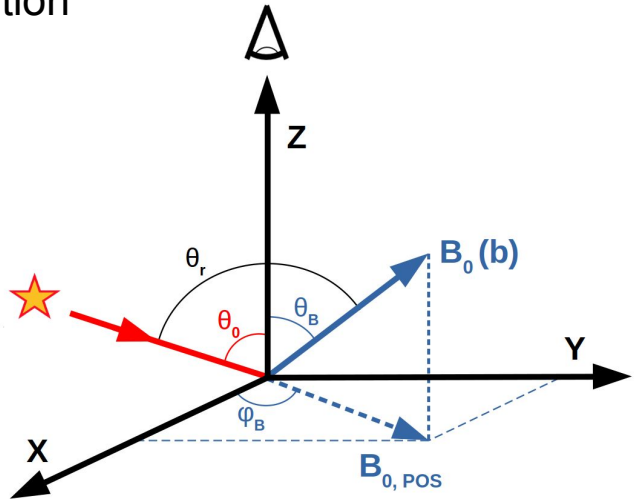

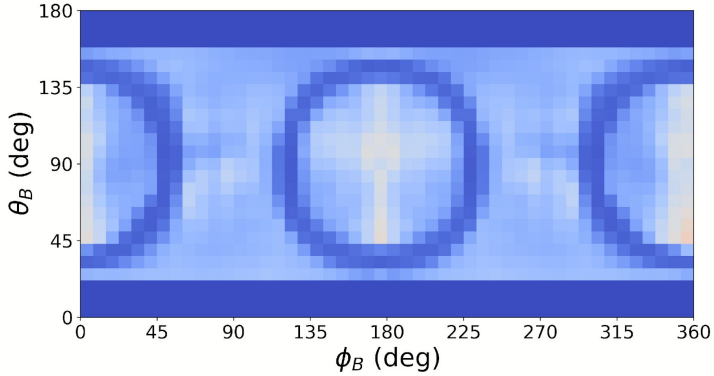

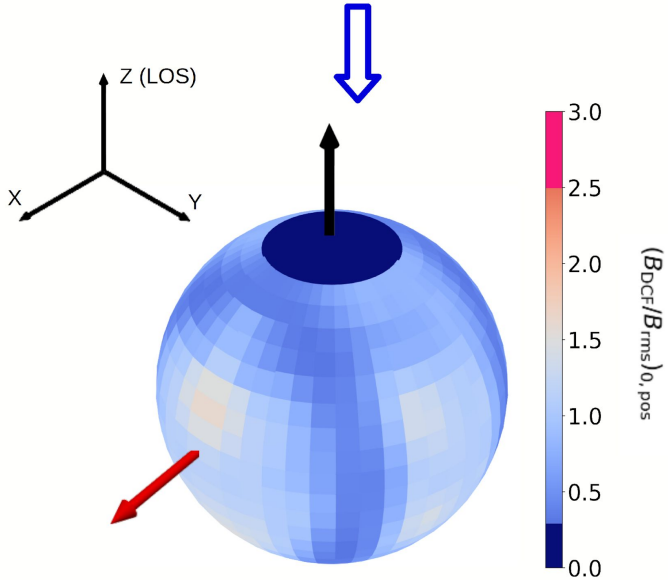

Testing DCF with GSA

Sample results

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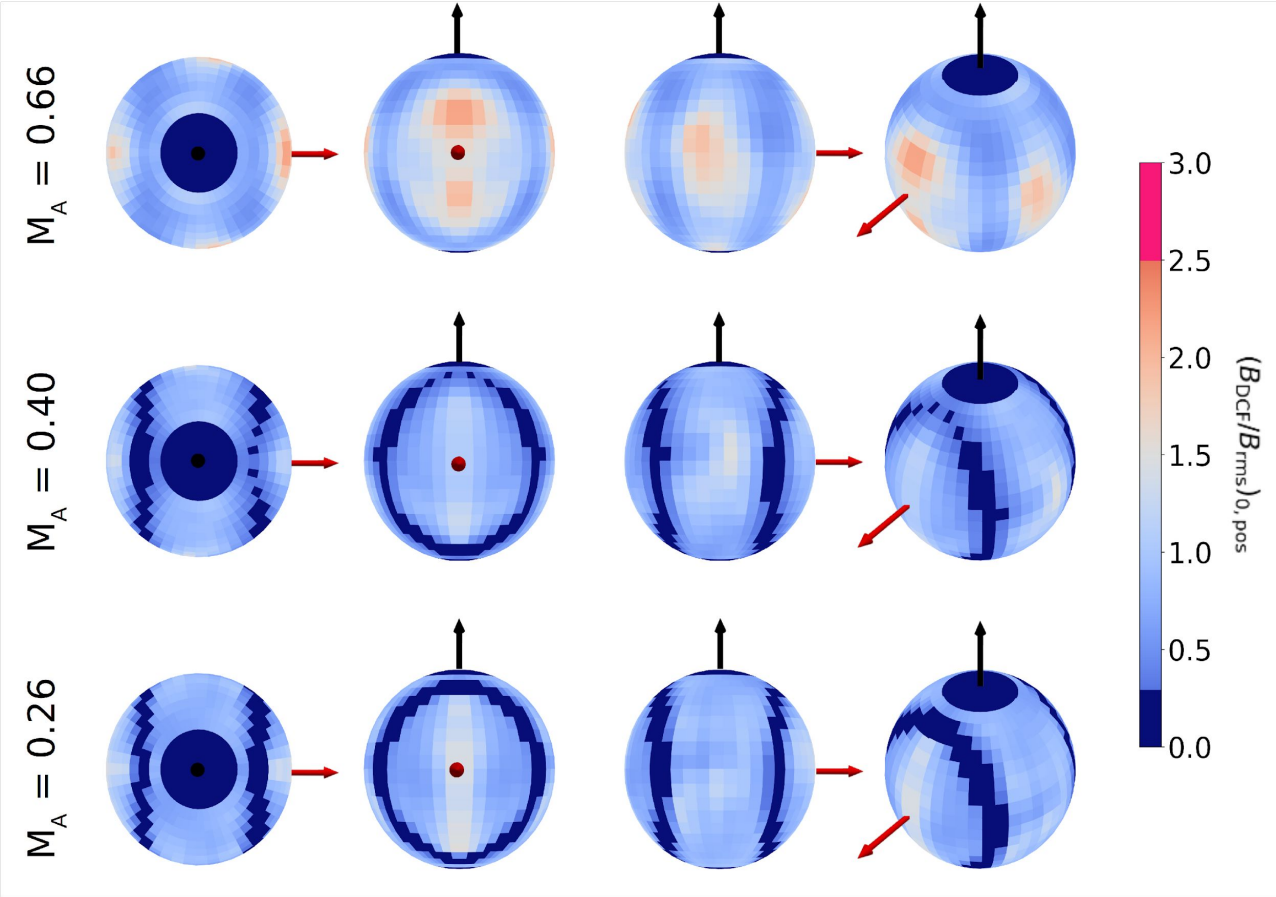
- Sample results from 3 simulations
- Radiation field direction along x-axis (θ_0 = 90°)



Mean B-field

Radiation field

Results

DCF comparison: GSA vs dust

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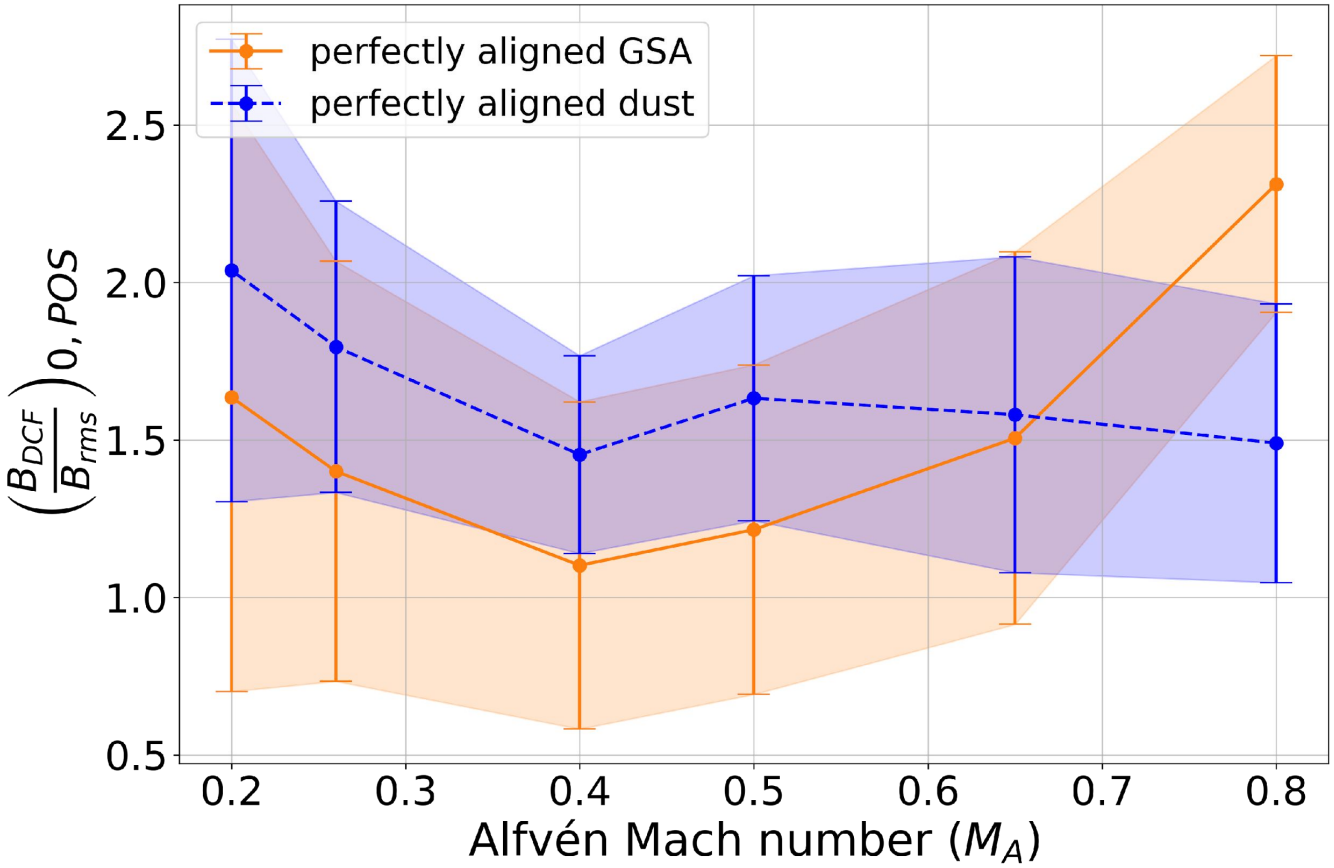
- Normalized DCF measured B-field vs M_A
- Values show averages over all mean field orientations



Results

Validity of CY16 method with GSA

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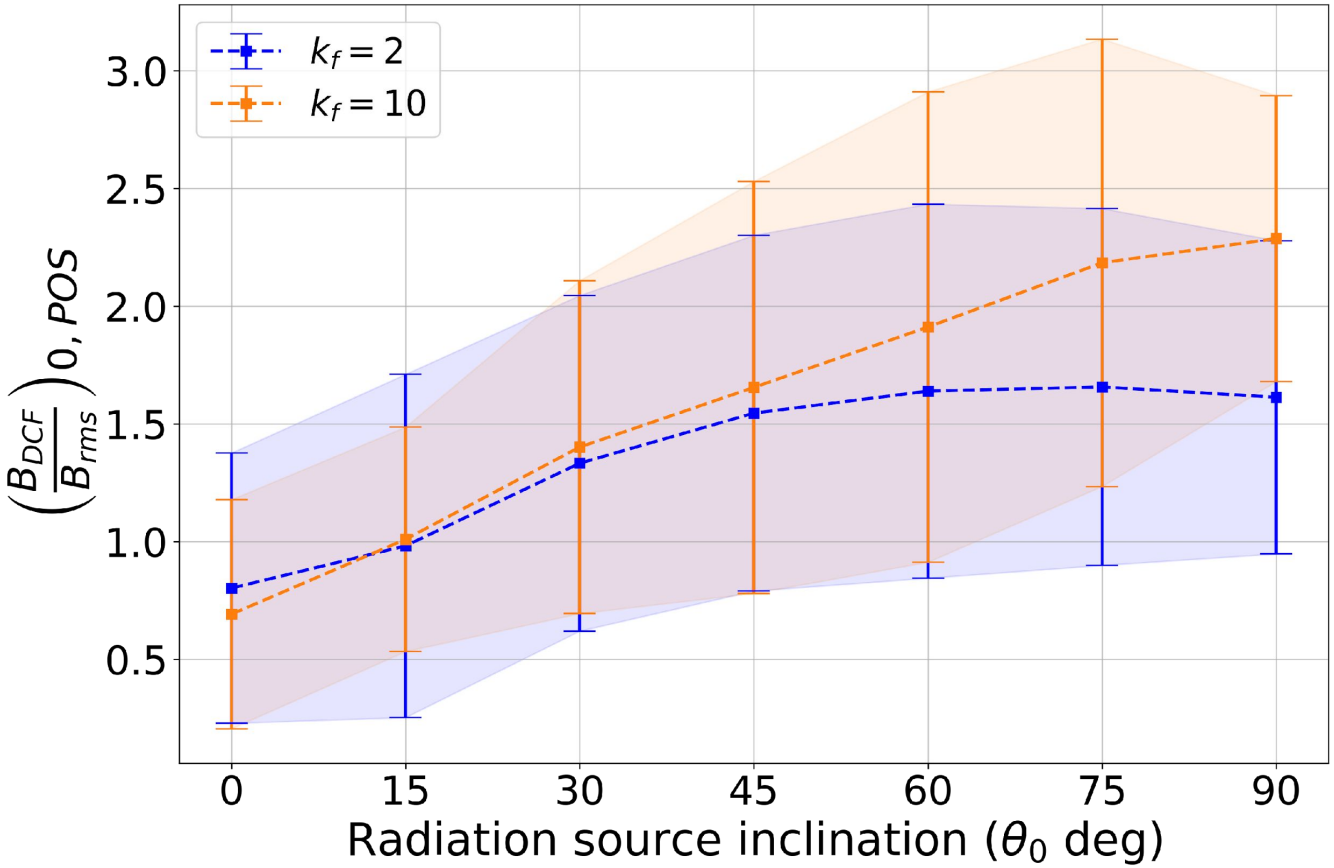
- Considering simulations with different driving scale
- Showing measured field vs radiation source inclination angle



Take-home points

- Polarization of atomic lines from GSA can be used with the DCF method to measure both LOS and POS observables
- DCF with GSA can measure B-field strengths as accurately as with dust.
- Cho & Yoo 2016 method is accurate regardless of the polarization mechanism and radiation source orientation.

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Thank you

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